

# Mathematical Foundation for AI Systems

## Session Agenda

Duration: 3 Hours

### Session Overview

This session introduces the core mathematical foundations that power modern AI systems — from classical Machine Learning through Deep Learning, Large Language Models, Retrieval-Augmented Generation (RAG), and Agentic AI. Learners progress from foundational probability and Combinations through vectors, matrices, calculus, similarity metrics, and optimization, connecting each concept directly to how real GenAI systems work — supported by beginner-friendly examples and a hands-on notebook.

### Learning Outcomes

- Understand how mathematics powers AI systems, from classical ML to modern GenAI
- Build intuition behind probability, conditional probability, and Bayes' Theorem
- Understand vectors, embeddings, and semantic space
- Perform basic matrix and tensor operations behind transformer models
- Interpret how LLMs use probability for next-token prediction
- Understand gradients and how AI systems learn from error
- Apply similarity metrics such as cosine similarity, used in RAG and semantic search

### Detailed Session Schedule

| Duration                                        | Topic / Content                                                                                                                                                 |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part 1 — Foundations of AI Mathematics (85 min) |                                                                                                                                                                 |
| 5 min                                           | <b>Why Math for AI:</b> Framing the session — AI as Math + Data + Code, and why this foundation matters before diving into ML/LLMs.                             |
| 15 min                                          | <b>AI Mathematics Mindset:</b> Thinking like a scientist, engineer, and decision-maker; permutations & combinations intuition through a simple example.         |
| 25 min                                          | <b>Probability Foundations:</b> What probability means, conditional probability, and Bayes' Theorem walked through with a natural-frequency tree example.       |
| 20 min                                          | <b>Distributions &amp; Sampling:</b> Binomial and Poisson intuition using simple, relatable scenarios — no heavy formulas.                                      |
| 20 min                                          | <b>Vectors, Embeddings &amp; Semantic Space:</b> Vector = a list of numbers; embedding = meaning as numbers; visualized with a toy 2D word map.                 |
| 10 min                                          | <i>Break</i>                                                                                                                                                    |
| Part 2 — Math Behind Modern AI Systems (85 min) |                                                                                                                                                                 |
| 15 min                                          | <b>Matrices, Tensors &amp; Transformer Operations:</b> Matrix as a grid of vectors; tensor as a generalization; how these connect to attention in transformers. |
| 15 min                                          | <b>Probability &amp; Token Prediction:</b> How LLMs predict the next word — worked example with a next-token probability table.                                 |
| 20 min                                          | <b>Calculus, Gradients &amp; Learning:</b> Slope-to-gradient-descent intuition using a simple $y = wx$ model with a hand-calculated gradient step.              |

| Duration | Topic / Content                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 15 min   | <b>Similarity Metrics &amp; Vector Search:</b> Cosine similarity explained, and why RAG systems retrieve the closest matching document. |
| 10 min   | <b>Optimization &amp; AI Agent Decisioning:</b> How AI agents choose actions using utility scores and reward-based optimization.        |
| 10 min   | <b>Wrap-up + Hands-on Notebook Walkthrough:</b> Recap of key concepts and a guided live-coding walkthrough of the notebook examples.    |